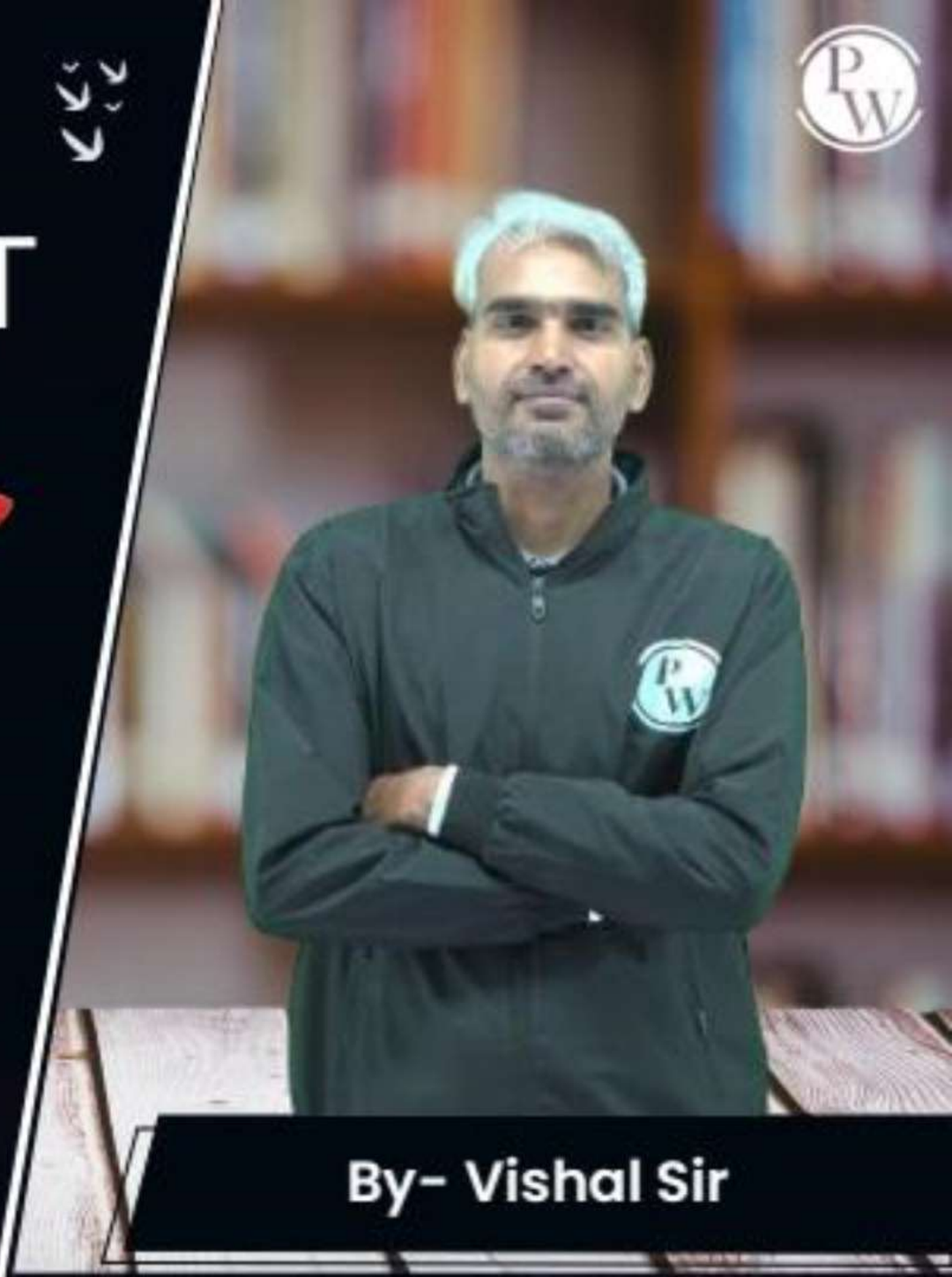
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 06



By- Vishal Sir

Recap of Previous Lecture



- **Topic** Closure of an attribute set
- **Topic** Super key
- **Topic** Candidate key
- **Topic** Identification of candidate key w.r.t. FD set

Slide

Topics to be Covered



- Topic: Membership test
- Topic: Relationship between two FD sets
- Topic: FD set of a sub-relation
- Topic: Minimal Cover (Canonical Cover)

Slide

HW

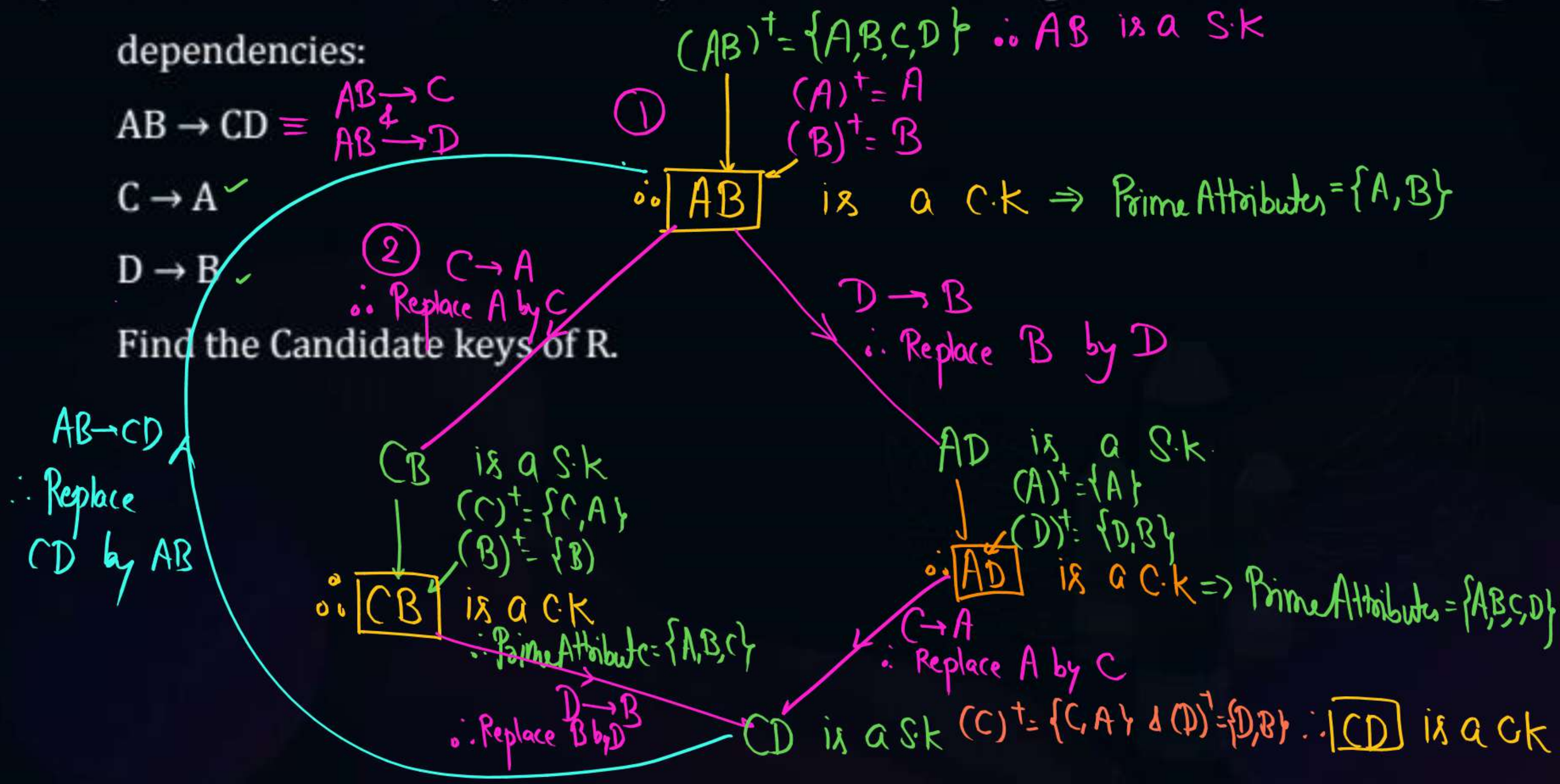
#Q. Assume a relation R (A, B, C, D) that has the following functional dependencies:

$AB \rightarrow CD \equiv AB \rightarrow C$, $AB \rightarrow D$

$C \rightarrow A$ ✓

$D \rightarrow B$ ✓

Find the Candidate keys of R.



H.W.

#Q. Assume a relation R (A, B, C, D, E, H) that has the following functional dependencies:

$A \rightarrow B$

$BC \rightarrow D$

$E \rightarrow C$

$D \rightarrow A$

Find the Candidate keys of R.

E & H are Essential attributes for C.K.

$(EH)^+ = \{E, H, C\}$ Not all attributes ∴ EH is not a key

$A \rightarrow B$ If we replace B by A, then we get AEH which is already a C.K.

$(AEH)^+ = \{A, E, H, B, C, D\}$ all attributes ∴ AEH is a S.K.

No proper subset of {A, E, H} is a S.K

∴ AEH is a C.K.

Prime Attributes = {A, E, H}

$D \rightarrow A$ (P.A)

∴ Replace A by D

DEH is a S.K

No proper subset of {D, E, H} is a S.K. ∴ DEH is a C.K ⇒ Prime Attributes = {A, E, H, D}

$BC \rightarrow D$

∴ Replace D by BC

BCEH is a S.K.

Check for C.K

$(BFH)^+ = \{B, E, H, C, D, A\}$

$(CEH)^+ = \{C, E, H\}$

BEH is a C.K ∴ BCEH can not be a C.K

No proper subset of BEH is a key

∴ BEH is a C.K.

⇒ Prime Attributes = {A, E, H, D, B}



Topic : Membership test

- Membership test is used to check whether a given FD is a member of given FD set or not.

- To check whether X→Y is a member of FD set F or not
  (i.e., $F \models X \rightarrow Y$ or not) ← F implies X→Y or not

  We first obtain X^+ (closure of X) w.r.t. FD set F.

  If $Y \in X^+$, then X→Y is a member of FD set F
  otherwise not a member of FD set F

- Given FD set F and a functional dependency $X \rightarrow Y$

* How do we check whether $X \rightarrow Y$ is member of FD set F or not { FD $X \rightarrow Y$ is implied by F or not }

eg: Let FD set $F = \{A \rightarrow B, B \rightarrow C\}$

Check whether FD $A \rightarrow C$ is a member of F or not?

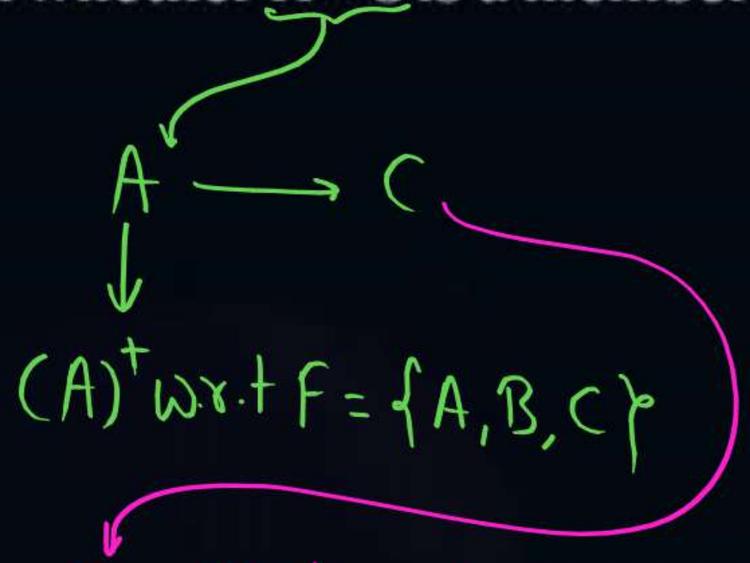
* By transitivity we know that $A \rightarrow C$ is a member of F.

$(A)^+$ w.r.t $F = \{A, B, C\}$

does $C \in (A)^+$ w.r.t. F ?

#Q. Let FD set $F = \{ A \rightarrow B, B \rightarrow C \}$

Check whether $A \rightarrow C$ is a member of F or not?

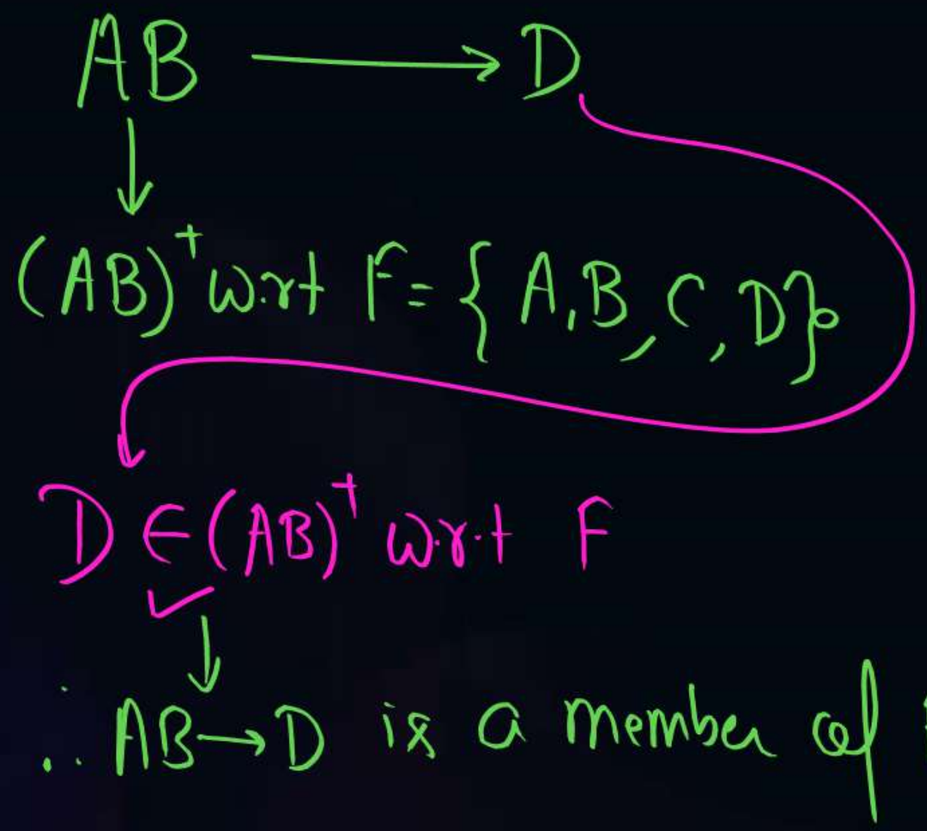


$C \in (A)^+$ w.r.t. F

$\therefore A \rightarrow C$ is a Member of F

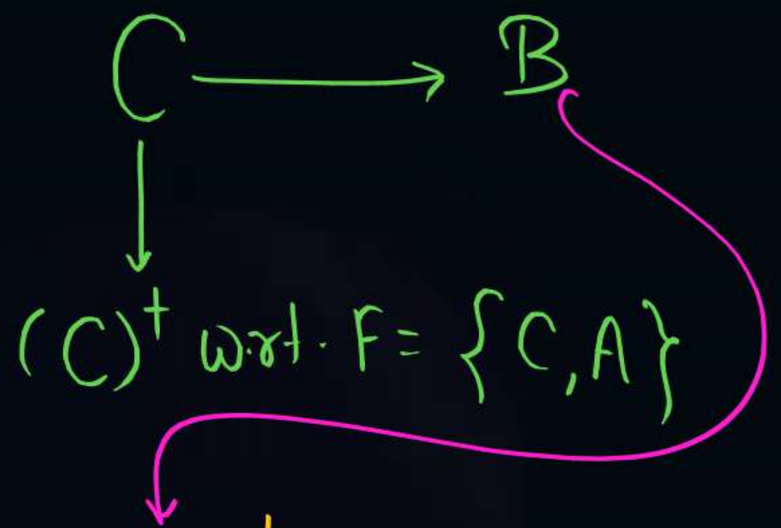
#Q. Let FD set F = { AB→C, BC→D}

Check whether AB→D is a member of F or not?



#Q. Let FD set F = { AB→C, C→A}

Check whether C→B is a member of F or not?



Note:- If FD $X \longrightarrow Y$ is a member of FD set F, then we say that $X \longrightarrow Y$ is implied by F.

* Let F & G are two FD sets.

① If Every FD of set F are member of FD set G, then we can say $F \subseteq G$

② If Every FD of set G are member of FD set F, then we can say $G \subseteq F$

Note: If $F \subseteq G$ as well as $G \subseteq F$, then $F = G$



Topic : Relationship between two FD sets

- Let F and G are any two FD sets.

- If all the FDs of FD set F are member of FD set G, then $F \subseteq G$
  {i.e FD set G Covers FD set F}

- If all the FDs of FD set G are member of FD set F, then $G \subseteq F$
  {i.e. FD set F Covers FD set G}

- If both $F \subseteq G$ and $G \subseteq F$ are true, then F = G
  {i.e F Covers G
  &
  G Covers F}

#Q. Consider the following FD sets

$F1 = \{A \to B, B \to C, AB \to D\}$ and $F2 = \{A \to B, B \to C, A \to C, A \to D\}$

Find the relationship between FD sets F1 and F2

- Check whether F_1 Covers F_2 or not?
  {i.e. check if all FDs of F_2 are member of F_1 or not?}

FD of set F_2	To check if Member of f_1 or not, obtain Closure w.r.t F_1	FD is a Member or not
$A \to B$	$(A)^+$ w.r.t $F_1 = \{A, B, C, D\}$	$B \in (A)^+$ ∴ Member
$B \to C$	$(B)^+$ w.r.t $F_1 = \{B, C\}$	$C \in (B)^+$ ∴ Member
$A \to C$	$(A)^+$ w.r.t $F_1 = \{A, B, C, D\}$	$C \in (A)^+$ ∴ Member
$A \to D$	$(A)^+$ w.r.t $F_1 = \{A, B, C, D\}$	$D \in (A)^+$ ∴ Member

Does F_1 Covers F_2

All FDs of F_2 are Member of F_1
i.e. F_1 Covers F_2

Hence $\boxed{F_2 \subseteq F_1}$ —eq^n (1)

ⓐ $F_1 \subseteq F_2$

ⓑ $F_2 \subseteq F_1$

ⓒ $F_1 = F_2$

ⓓ None

#Q. Consider the following FD sets

$F1 = \{A \rightarrow B, B \rightarrow C, AB \rightarrow D\}$ and $F2 = \{A \rightarrow B, B \rightarrow C, A \rightarrow C, A \rightarrow D\}$

Find the relationship between FD sets F1 and F2

- Check whether F_2 Covers F_1 or not..?
  {i.e. Check if all FDs of F_1 are member of F_2 or not?}

FD of set F_1	To check if Member of F_2 or not, obtain Closure w.r.t F_2	FD is a Member or not	Does F_2 Covers F_1.
$A \rightarrow B$	$(A)^+$ w.r.t $F_2 = \{A, B, C, D\}$	$B \in (A)^+$ Member	All FDs of F_1 are member of F_2 ∴ F_2 Covers F_1 Hence $F_1 \subseteq F_2$ — eqn. ②
$B \rightarrow C$	$(B)^+ = \{B, C\}$	$C \in (B)^+$ Member	
$AB \rightarrow D$	$(AB)^+ = \{A, B, C, D\}$	$D \in (AB)^+$ Member	

ⓐ $F_1 \subseteq F_2$

ⓑ $F_2 \subseteq F_1$

ⓒ $F_1 = F_2$

ⓓ None

By eqn ① & ②

$$F_1 = F_2$$

#Q. Consider the following FD sets

F1 = {A→B, B→C,A→C} and F2 = {A→B, B→C, A→D}

Find the relationship between FD sets F1 and F2

$F_1 = \{ A \to B$ ✓ Member of F_2

$B \to C$ ✓ Member of F_2

$A \to C$ ✓ Member of F_2

$\}$

$\therefore F_2$ Covers F_1

$F_1 \subseteq F_2$ — eq^n (1)

$F_2 = \{ A \to B$ ✓ Member of F_1

$B \to C$ ✓ Member of F_1

$A \to D$ $(A)^+$ w.r.t $F_1 = \{A, B, C\}$: $D \notin (A)^+$

$\therefore A \to D$ is not a Member of F_1

$\}$

$\therefore F_1$ does not Cover F_2

i.e $F_2 \not\subseteq F_1$

#Q. Consider the following FD sets

$F1 = \{A \rightarrow B, AB \rightarrow C, D \rightarrow AC, D \rightarrow E\}$ and

$F2 = \{A \rightarrow BC, D \rightarrow AE\}$

Find the relationship between FD sets F1 and F2

(a) $F_1 \subseteq F_2$

(b) $F_2 \subseteq F_1$

(c) $F_1 = F_2$

(d) None

$F_1 = \{ A \rightarrow B$ ✓ Member of F_2

$AB \rightarrow C$ ✓ Member of F_2

$D \rightarrow AC$ $(D)^+$ wrt $F_2 = \{D, A, E, B, C\}$ ∴ Member of F_2

$D \rightarrow E$ ✓ Member $\}$

∴ all FDs of F_1 are Member of F_2

∴ $\boxed{F_1 \subseteq F_2}$ – eq^n ①

$F_2 = \{ A \rightarrow BC$ ✓ $(A)^+$ wrt $F_1 = \{A, B, C\}$ ∴ Member

$D \rightarrow AE$ $(D)^+$ wrt $F_1 = \{D, A, C, E, B\}$

∴ Member ✓ $\}$

All FDs of F_2 are Member of F_1

∴ $\boxed{F_2 \subseteq F_1}$ – eq^n ②

By ① & ② $F_1 = F_2$

FD set of a sub-relation :-

Let $R(A, B, C, D, E, F)$ be the relational schema with FD set F,

If R_1 is a sub-relation of relation R {let with attributes A, C, E of relation R} i.e. $R_1(A, C, E)$, then can we identify the FDs that holds true in sub-relation $R_1(A, C, E)$



Topic : FD set of a sub-relation

- Let R be a relation with FD set F. We can use the concept of membership test to obtain the FD set of a sub-relation of relation R.

#Q. Consider a relational schema R(A,B,C,D,E) with FD set

$F = \{A \rightarrow B, AB \rightarrow C, D \rightarrow AC, D \rightarrow E\}$

Find the FD set F1 for sub-relation R1(A,B,E) of R(A,B,C,D,E).

Also find candidate keys for the sub-relation R1(A,B,E).

→ To identify the Candidate key of any relation, We need functional dependencies which holds true in that relation.

#Q. Consider a relational schema R(A,B,C,D,E) with FD set

$F = \{A \rightarrow B,\ AB \rightarrow C,\ D \rightarrow AC,\ D \rightarrow E\}$

Find the FD set F1 for sub-relation R1(A,B,E) of R(A,B,C,D,E).

Also find candidate keys for the sub-relation R1(A,B,E).

R_1 is a relation with attributes A, B, E
the dependency between A, B, E in sub-relation R_1 will follow from the dependencies that holds true between A, B, & E in relation R.

Non-trivial FDs in relation R_1 may be of the type

A → Something
B → Something
E → Something
AB → Something
AE → Something
BE → Something

#Q. Consider a relational schema R(A,B,C,D,E) with FD set

$F = \{A \rightarrow B,\ AB \rightarrow C,\ D \rightarrow AC,\ D \rightarrow E\}$

Find the FD set F1 for sub-relation R1(A,B,E) of R(A,B,C,D,E).

Also find candidate keys for the sub-relation R1(A,B,E).

$(A)^+$ w.r.t $F = \{A, B, C\}$

i.e. in relation R, A can determine A, B & C

i.e. $A \rightarrow ABC \Rightarrow A \rightarrow BC$

↓

in Subrelation R_1 'C' is not present

∴ w.r.t. R_1 $A \rightarrow B$ is a Valid FD

Non-trivial FDs in relation R_1 may be of the type

$A \rightarrow$ Something
$B \rightarrow$ Something
$E \rightarrow$ Something
$AB \rightarrow$ Something
$AE \rightarrow$ Something
$BE \rightarrow$ Something

#Q. Consider a relational schema R(A,B,C,D,E) with FD set

$F = \{A \to B,\ AB \to C,\ D \to AC,\ D \to E\}$

Find the FD set F1 for sub-relation R1(A,B,E) of R(A,B,C,D,E).

Also find candidate keys for the sub-relation R1(A,B,E).

$R_1(A,B,E)$

$(A)^+$ w.r.t $F = \{A,B,C\}$
i.e. $A \to ABC \Rightarrow A \to BC$ holds in R
$\therefore \boxed{A \to B}$ will hold in R_1

$(B)^+$ w.r.t $F = \{B\}$
i.e. $B \to B$ holds in R,
Nothing Non-trivial w.r.t B in R

$(E)^+$ w.r.t $F = \{E\}$
i.e. $E \to E$ in R
trivial.

$(AB)^+$ w.r.t $F = \{A,B,C\}$
i.e. $AB \to ABC$ in R
$AB \to C$ in R
not in R_1

$(AE)^+$ w.r.t $F = \{A,B,C,E\}$
i.e. $AE \to ABCE$ in R (trivial)
$AE \to BC$ not in $R_1 \Rightarrow \therefore \boxed{AE \to B}$ holds in R_1

$(BE)^+$ w.r.t $F = \{B,E\}$
$\therefore BE \to BE$ in R
trivial

∴ FDs w.r.t sub-relation R_1 are

$A \rightarrow B$ & $AE \rightarrow B$

∴ FD set $F_1 = \{A \rightarrow B, AE \rightarrow B\}$ w.r.t. $R_1(A,B,E)$

C.k. for $R_1(A,B,E)$ will be = (AE)

Prime Attribute = $\{A, E\}$

#Q. Consider a relational schema R(A,B,C,D,E,F) with FD set

$F = \{AB \to C,\ B \to D,\ BC \to A,\ D \to EF\}$

Find the FD set F1 for sub-relation R1(A,B,C,D) of R(A,B,C,D,E,F).

Also find candidate keys for the sub-relation R1(A,B,C,D).

FD set of $R_1(A,B,C,D)$

i.e. $F_1 = \{B \to D,\ AB \to CD,\ BC \to AD,\ ABC \to D,\ ABD \to C,\ BCD \to A\}$

Candidate keys of R_1 = AB & BC

w.r.t $R_1(A,B,C,D)$

$(A)^+$ w.r.t. $F = \{A\}$ $\quad A \to A$ ✗

$(B)^+$ w.r.t. $F = \{B,D,E,F\}$

$B \to D$ $\quad B \to BDEF \Rightarrow B \to DEF$ (trivial; Not in R_1)

$(C)^+$ w.r.t. $F = \{C\}$

$C \to \phi$

$(D)^+$ w.r.t. $F = \{D,E,F\}$

$D \to DEF \Rightarrow D \to EF$ (trivial; Not in R_1)

$(AB)^+ = \{A,B,C,D,E,F\}$ (trivial; Not in R_1)

$AB \to CD$

$(AC)^+ = \{A,C\}$ (trivial)

$(AD)^+ = \{A,D,E,F\}$ (trivial; Not in R_1)

$(BC)^+ = \{B,C,A,D,E,F\}$ (trivial; Not in R_1)

$BC \to AD$

$(BD)^+ = \{B,D,E,F\}$ (trivial; Not in R_1)

$(CD)^+ = \{C,D,E,F\}$ (R_1)

$(ABC)^+ = \{A,B,C,D,E,F\}$

$ABC \to D$

$(ABD)^+ = \{A,B,D,C,E,F\}$

$ABD \to C$

$(ACD)^+ = \{A,C,D,E,F\}$

$(BCD)^+ = \{B,C,D,A,E,F\}$

$BCD \to A$



Topic : Minimal cover (Canonical cover)

- Minimal cover of canonical cover of FD set F is a set of functional dependencies (F_m)
  such that,
  - $F_m = F$ and
  - F_m does not contain any redundant FD or extraneous attribute at either side of any FD of F_m

#e.g., Consider the FD set

$F = \{A \rightarrow B,\ B \rightarrow C,\ A \rightarrow C\}$

#e.g., Consider the FD set

$F = \{AB \rightarrow C,\ A \rightarrow B\}$



Topic : Testing if an Attribute is Extraneous

Consider a set F of functional dependencies and functional dependency $\alpha \rightarrow \beta$ in F.

To test if attribute $A \in \alpha$ is extraneous in α (i.e., Any extraneous attribute in LHS of FD)

1. compute $(\{\alpha\} - \{A\})^+$ using the dependencies in F.
2. check if $(\{\alpha\} - A)^+$ contains A; if it does then, A is extraneous.

To test if attribute $A \in \beta$ is extraneous in β (i.e., Any extraneous attribute in RHS of FD)

1. compute α^+ using only the dependencies in $F' = (F - \{\alpha \rightarrow \beta\}) \cup \{\alpha \rightarrow (\beta - A)\}$.
2. check that α^+ contains A; if it does then, A is extraneous



Topic : Procedure to obtain minimal cover of FD set

1. Simplify RHS of all FDs (i.e., split the FDs such that RHS contain exactly one attribute)
2. For all FDs find redundant (extraneous) attribute in LHS.
3. Eliminate all redundant FDs
4. Apply Union if needed
5. The result is minimal Cover

#e.g., Consider the following FD set

F ={AC →G

D→EG

BC→D

CG→BD

ACD→B

CE→AG

}

Find minimal cover of F.



Topic : NOTE

Minimal cover of FD set F need not be unique, but all minimal cover are logically equivalent.



2 mins Summary

- **Topic** Membership test
- **Topic** Relationship between two FD sets
- **Topic** FD set of a sub-relation
- **Topic** Minimal Cover (Canonical Cover)

$F_1 \subseteq F_2$
$F_2 \subseteq F_1$
$F_1 = F_2$
none

THANK - YOU